

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
II SEMESTER 2012-2013

CE C212

Transport Phenomena
Comprehensive Examination

Dated: 2.05.2013

Max. Marks: 80
minutes

Duration: 180

Question paper contains two parts.

Answer all questions.

Answer all parts of a question together.

PART A (Closed Book)

1. Select the most appropriate alternative:

(10x1=10)

i). Practical fluid possess

- a) viscosity b) surface tension
c) compressibility d) all the above

ii). Water belongs to

- a) Newtonian fluids b) non Newtonian fluids
c) dynamic viscosity d) none of these

iii). The intensity of pressure developed by surface tension of 0.075 N/m in a droplet of water of 0.075 mm diameter is

- a) 0.8 N/cm² b) 0.6 N/cm²
c) 0.4 N/cm² d) 0.5 N/cm²

iv). If cohesion between the molecules of a fluid is more than adhesion between the fluid and glass, the free level of fluid in a dipped glass tube will be

- a) higher than the surface of liquid b) same as the surface of liquid
c) lower than the surface of liquid d) none of these

v). Gauge pressure is

- a) absolute pressure – atmospheric pressure b) absolute pressure + atmospheric pressure
c) atmospheric pressure – absolute pressure d) none of these

vi). On an inclined plane, centre of pressure is located

- a) at the centroid b) above the centroid
c) below the centroid d) anywhere

vii). Bernoulli's equation assumes that

- a) fluid is non-viscous
- b) fluid is homogeneous
- c) flow is steady
- d) flow is along the stream line
- e) all the above

viii). Centre of buoyancy is

- a) centroid of the floating body
- b) centroid of the fluid displaced
- c) centre of pressure of displaced liquid
- d) none of these

ix). The acceleration f required to accelerate a rectangular tank containing water horizontally so that the slope

of its free surface is 45° , is

- a) $g/2$
- b) g
- c) $2g$
- d) $2.5g$
- e) $3g$

x). The shape of fire hose nozzle is generally kept

- a) divergent
- b) convergent
- c) convergent divergent
- d) cylindrical

2. (a) Prove that the normal stress at a point in a static fluid is independent of direction.

(b) Prove that pressure intensity at any point in a static mass of fluid does not vary in x and y direction.

(c) Derive Euler's equation.

(d) What do you understand by Stable, Unstable and Neutral equilibrium?

(e) Derive an equation for total hydrostatic force on inclined surface.

(3x5=15)

3. (a) Show that vorticity is equal to twice the rotation component about an axis perpendicular to the plane in

which area is lying.

(b) Consider the two-dimensional flow field in which $u = Axy$ and $v = By^2$, where $A = 1 \text{ m}^{-1}\text{s}^{-1}$ and $B =$

$-1/2 \text{ m}^{-1}\text{s}^{-1}$ and the coordinates are measured in meters. Show that the velocity field represents a possible

incompressible flow. Determine the rotation at point $(x,y) = (1,1)$. Evaluate the circulation about the

"curve" bounded by $y=0$, $x=1$, $y=1$ and $x=0$.

(c) For an orifice, show that $C_d = C_c \times C_v$.

(2+6+2)

PART B (Open Book)

4. (a) An incompressible fluid flows past a flat plate, as in the figure below, with a uniform inlet profile and a polynomial exit profile

$$V_x = V_0 \left(\frac{3\eta - \eta^3}{2} \right), \text{ where } \eta = \frac{y}{\delta}$$

Compute the volume flow Q across the top surface of the control volume. The plate has width b into the paper.

(10)

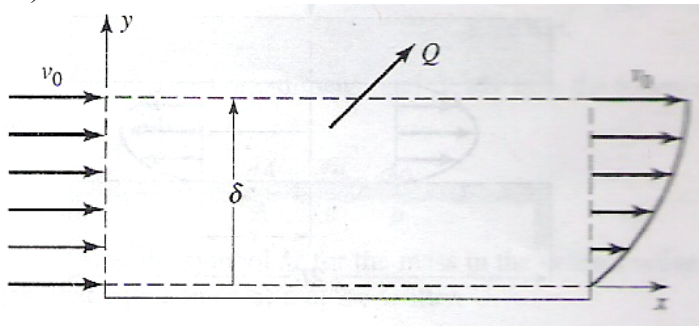


Figure Q 4(a)

- (b) When A free jet of water with constant cross – section area 0.005 m^2 is deflected by a hinged plate of

length $L = 2 \text{ m}$ supported by a spring constant $k = 1 \text{ N/m}$ and uncompressed length $x_0 = 1 \text{ m}$. Find the

deflection angle Θ as a function of jet speed V . What jet speed has a deflection of 10° ?

(15)

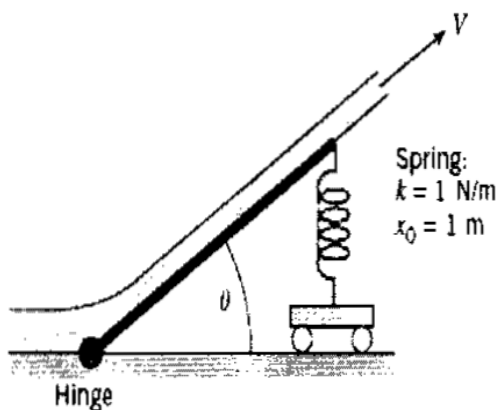


Figure Q 4(b)

5. (a) A flat plate orifice of 50 mm diameter is located at the end of a 100 mm diameter pipe. Water flows through the pipe and orifice at $0.57 \text{ m}^3/\text{s}$. The diameter of the water jet downstream from the orifice is 38 mm . Calculate the external force required to hold the orifice in place. Neglect the friction

on the pipe wall.
(10)

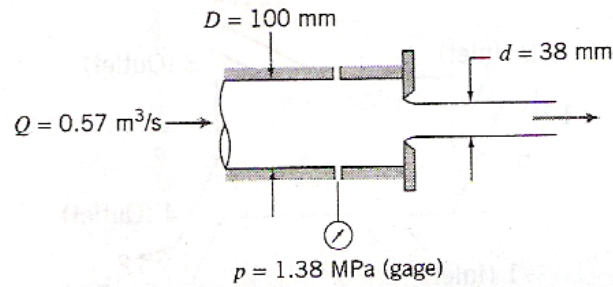


Figure Q 5(a)

(b) In a smooth pipe of uniform diameter 25 cm a pressure of 50 kPa was observed at section 1 which was at elevation 10 m. At another section 2 at elevation 12 m the pressure was 20 kPa and the Velocity was 1.25 m/s. Determine the direction of flow and the head loss between these two sections.

(10)

Paper ends